NUMBER GUESSING GAME DEVELOPMENT USING PYTHON

AIM

To design and develop a simple Number Guessing Game using Python Tkinter. The program generates a random number between 1 and 20 and allows the player to guess the number through a graphical interface. It provides feedback, counts attempts, and includes restart and exit options.

PROCEDURE

1. Start the program and import the required Python libraries: tkinter and random.
2. Generate a random number between 1 and 20 using the random module.
3. Create the main game window using Tkinter and set its title.
4. Add a title, instructions, number-entry box, Guess button, and result message.
5. Read the number entered by the player and validate the input.
6. Compare the guess with the secret number.
7. Display "Too Low!" if the guess is smaller or "Too High!" if the guess is larger.
8. When the correct number is entered, display a winning message and the number of attempts.
9. Add a Restart button to generate a new number and reset the attempts.
10. Add an Exit button to close the game window.
11. Run the program in Python IDLE and test the game controls.

PROGRAM CODE

```
import tkinter as tk
import random

number = random.randint(1, 20)
attempts = 0

def check_guess():
    global attempts

    try:
        guess = int(entry.get())
    except ValueError:
        result.config(text="Enter a valid number!")
        return

    if guess < 1 or guess > 20:
        result.config(text="Enter a number from 1 to 20!")
        return

    attempts += 1

    if guess < number:
        result.config(text="Too Low! Try Again.")
    elif guess > number:
        result.config(text="Too High! Try Again.")
    else:
        result.config(text="You Won! Attempts: " + str(attempts))

def restart_game():
```

```
    global number, attempts
    number = random.randint(1, 20)
    attempts = 0
    entry.delete(0, tk.END)
    result.config(text="Guess a number between 1 and 20")

window = tk.Tk()
window.title("Number Guessing Game")
window.geometry("400x350")
window.config(bg="lightblue")

tk.Label(
    window,
    text="NUMBER GUESSING GAME",
    font=("Arial", 16, "bold"),
    bg="lightblue"
).pack(pady=20)

tk.Label(
    window,
    text="Guess a number between 1 and 20",
    bg="lightblue"
).pack()

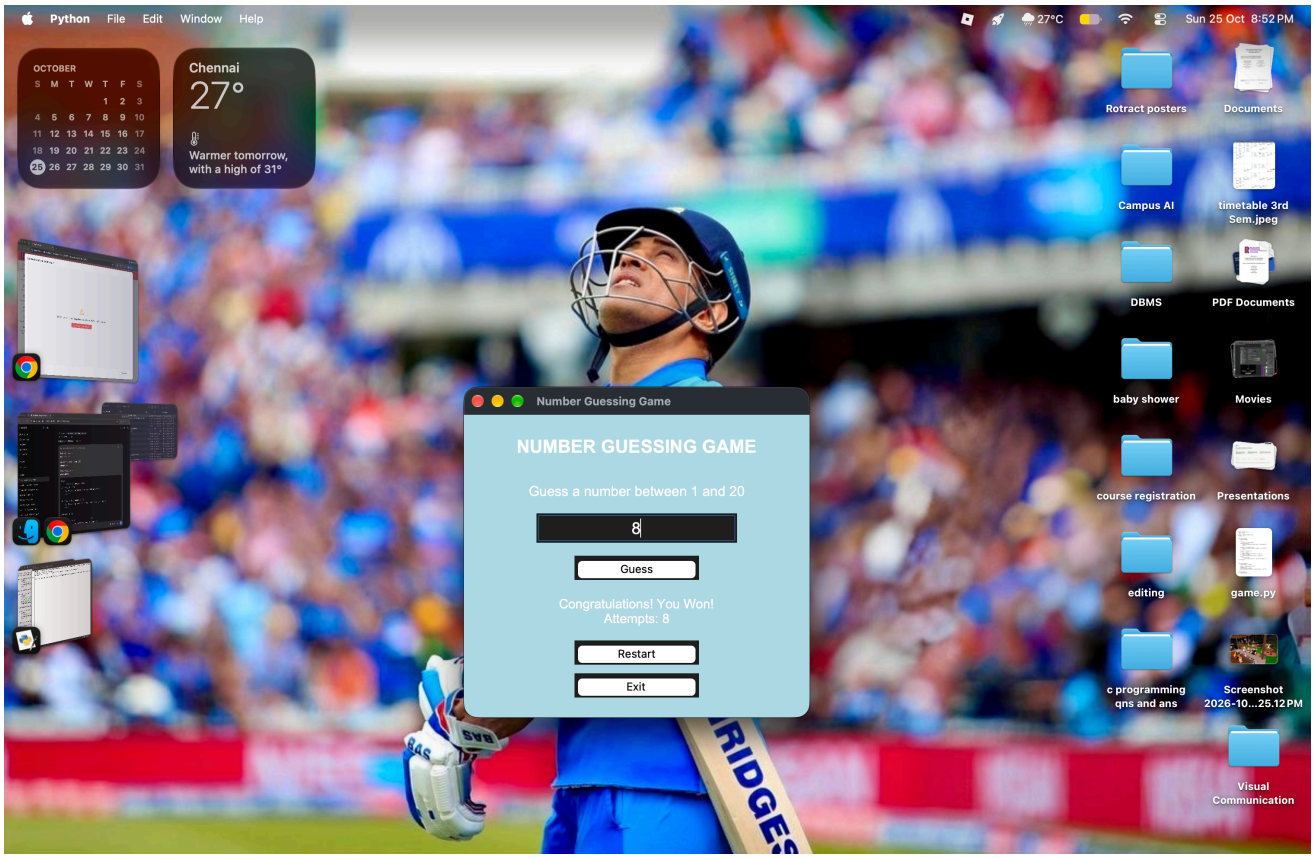
entry = tk.Entry(window, font=("Arial", 14), justify="center")
entry.pack(pady=10)

tk.Button(window, text="Guess", command=check_guess).pack(pady=5)

result = tk.Label(
    window,
    text="Enter your guess!",
    bg="lightblue"
)
result.pack(pady=10)

tk.Button(window, text="Restart", command=restart_game).pack(pady=5)
tk.Button(window, text="Exit", command=window.destroy).pack(pady=5)

window.mainloop()
```



OUTPUT

The game window displays the title, instructions, a number input box, a Guess button, feedback messages, a Restart button, and an Exit button. When the player guesses correctly, the game displays a winning message and the number of attempts.

RESULT

Thus, a Number Guessing Game was developed successfully using Python Tkinter. The game generates a random number, accepts user guesses, provides feedback, counts attempts, and offers restart and exit controls. The program demonstrates GUI design, conditional statements, functions, input validation, and random number generation.